

JEE MAIN REVISION MASTER: DUAL NATURE OF RADIATION SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 207 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Photoelectric Effect & Einstein's Equation	126 PYQs	Photoelectric
Very High	De Broglie Hypothesis & Matter Waves	81 PYQs	Matter Waves

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Photoelectric Effect & Einstein's Equation	$E = h\nu = \phi_0 + K_{\text{max}}$, $K_{\text{max}} = eV_s = h(\nu - \nu_0) = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$
De Broglie Hypothesis & Matter Waves	$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}}$, $\lambda_{\text{electron}} \approx \frac{1.227}{\sqrt{V}} \text{ nm}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Photoelectric Effect & Einstein's Equation (Very High Yield — 126 PYQs)

Core Principle: Quantum nature of light, work function ϕ_0 , threshold frequency ν_0 , and stopping potential V_s dependence on frequency.

■■ **Caution / JEE Pitfall:** Stopping potential depends solely on FREQUENCY and work function; it is completely independent of intensity of incident light!

■ **Speed Trick:** Slope of stopping potential vs frequency curve is always universal: Slope = h/e .

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons /s . The region of an electromagnetic spectrum to wh..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "Light source having wavelength 331 nm is used to generate photo-electrons whose stopping potential is 0.2 V . The work function of the used ..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "Number of photons of equal energy emitted per second by a 6 mW laser source operating at 663 nm is _____. (Given : $h=6.63 \times 10^{-34}$..." [Ans: D]

3.2 De Broglie Hypothesis & Matter Waves (Very High Yield — 81 PYQs)

Core Principle: Dual wave-particle nature of moving particles, de Broglie wavelength under electrostatic acceleration V , and thermal gas particles.

■ **Caution / JEE Pitfall:** For neutral particles (neutrons), de Broglie wavelength is $\lambda = \frac{h}{\sqrt{3mkT}}$, where k is Boltzmann constant.

■ **Speed Trick:** For proton vs alpha particle accelerated through same voltage V : $(\lambda_p / \lambda_\alpha) = \sqrt{(m_\alpha q_\alpha / m_p q_p)} = \sqrt{(4 \times 2)} = 2\sqrt{2}$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (Numerical):** "The ratio of de Broglie wavelength of a deuteron with kinetic energy E to that of an alpha particle with kinetic energy $2E$, is $n:1$. $T...$ " **[Ans: 2]**
- **[2026 Evening Shift] Q#2 (MCQ):** "The de Broglie wavelength associated with an electron accelerated through a potential difference V is $\lambda_e(e)$ and the de Broglie wavelength..." **[Ans: A]**
- **[2026 Evening Shift] Q#2 (Numerical):** "A particle having electric charge $3 \times 10^{-19}C$ and mass $6 \times 10^{-27}kg$ is accelerated by applying an electric potential of $1...$ " **[Ans: 10]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons/s. The region of an electromagnetic spectrum to which the emitted electromagnetic radiation belongs to ____ (Take $h = 6.6 \times 10^{-34}Js$ and $c = 3 \times 10^8 m/s$).

- (A) Microwave
- (B) Infrared
- (C) Visible
- (D) Ultraviolet **[Correct]**

Correct Answer: D

Solution Analysis: The power (P) of a light source is the total energy emitted per unit time. If the source emits n photons per second, and each photon has an energy E , then the power is: $P = n \times E$. The given power of the source is $P = 15kW = 15 \times 10^3W$ and the number of photons per second is $n = 2.5 \times 10^{22}s^{-1}$. $E = (P / n) \Rightarrow E = \frac{15 \times 10^3}{2.5 \times 10^{22}} \Rightarrow E = 6 \times 10^{-19}J$. The energy of a photon is related to its wavelength by the formula: $E = (hc / \lambda) \Rightarrow \lambda = (hc / E)$. Substituting the constants and energy, The PI

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Morning Shift (Q#1)

The ratio of de Broglie wavelength of a deuteron with kinetic energy E to that of an alpha particle with kinetic energy $2E$, is $n:1$. The value of n is ____ . (Assume mass of proton = mass of neutron) :

Correct Answer: 2

Solution Analysis: The de Broglie wavelength is given by : $\lambda = (h / p)$. Since kinetic energy $K = \frac{p^2}{2m}$, we can write momentum as $p = \sqrt{2mK}$. Therefore: $\lambda = \frac{h}{\sqrt{2mK}}$. Let m_p be the mass of a proton or neutron. Deuteron (d) consists of 1 proton and 1 neutron. The mass of deuteron is $m_d \approx 2m_p$. Kinetic Energy $K_d = E$. Alpha particle (α) consists of 2 protons and 2 neutrons. The mass of alpha particle is $m_\alpha \approx 4m_p$. Kinetic Energy $K_\alpha = 2E$. The ratio of wavelengths is : $\frac{\lambda_d}{\lambda_\alpha} = \frac{\frac{h}{\sqrt{2m_d K_d}}}{\frac{h}{\sqrt{2m_\alpha K_\alpha}}} = \frac{\sqrt{2m_\alpha K_\alpha}}{\sqrt{2m_d K_d}} = \frac{\sqrt{2 \times 4m_p \times 2E}}{\sqrt{2 \times 2m_p \times E}} = \frac{\sqrt{16m_p E}}{\sqrt{4m_p E}} = \frac{4}{2} = 2$.